TechPost ELEVATED

DC – AI – Routing – Switching – Security – Wireless – Automation

ACX7100/ACX7509 Validation: L2VPN

An HPE Juniper Networking Techpost

Suneesh Babu

2022-11-16

Contents

1 L2VPN Validation on ACX7000 **2**
1.1 Introduction . . . 2
1.2 Metro L2VPN Topology . . . 2
1.2.1 Base Configurations . . . 3
1.2.2 CE Bound Interface Schema . . . 4
1.2.3 L2VPN Verification . . . 4
1.2.4 Traffic Verification at Router at PE1 . . . 6
1.3 Generating Scale Configuration . . . 6
1.3.1 Jinja Configuration Template . . . 6
1.3.2 Configuration Parameters . . . 7

1 L2VPN Validation on ACX7000

Suneesh Babu - 11/16/2022

ACX7000 platform is tested with 8,000 Layer2 VPN Routing-instances for 99.9% line rate traffic.

1.1 Introduction

This is the third article in the ACX7k Metro Validation Series.

In this article, we validate the L2VPN feature scale support on the ACX7100-32C platform with 22.2R1 Junos-EVO image.

L2VPN is a BGP or LDP MPLS Enabled Solution to connect Customer Edge Devices (CEs) across a Provider Edge Network. RFC-6624 supports L2VPN using BGP for auto-discovery and Signalling. The PEs provide layer 2 bridged connectivity between CEs. The PEs of the network are connected via MPLS technology with the help of LDP, RSVP, SR-MPLS etc.

JUNOS-EVO supports L2VPN functionality with instance type “l2vpn”. 8,000 L2VPN Instances were tested on ACX7100-32C. The same scale is tested on ACX7100-48L as well as on ACX7509 Scale is not covered in this article (same features but different scale are expected on this platform).

Please note that despite sharing the same ACX moniker, the ACX7000 products are different products than ACX500/710/1000/1100/2100/2200/4000/5000/5400/6000. They are powered by different Packet Forwarding Engines (PFE), and support different feature sets and scales.

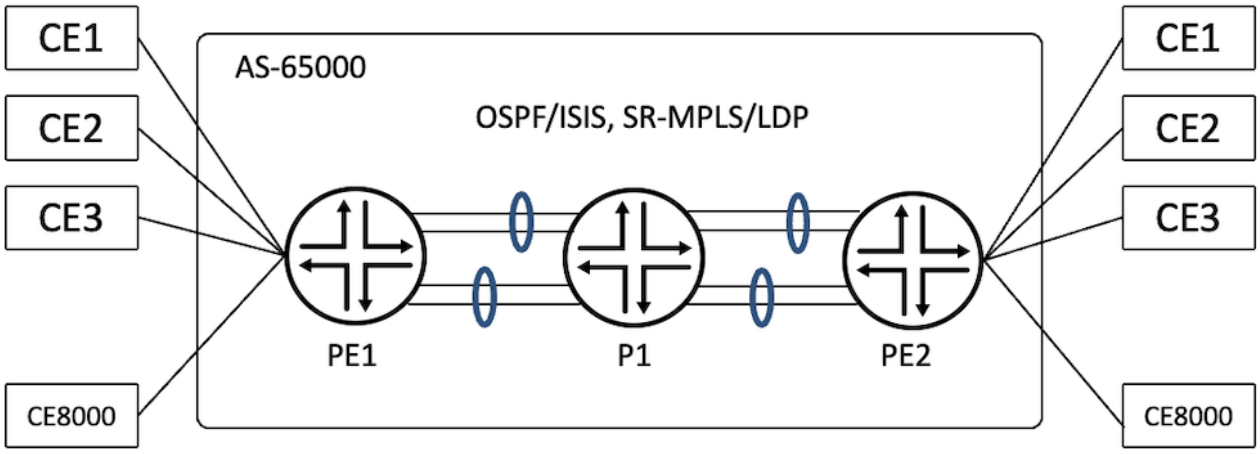
The Key Performance Indicator (KPI) tested are:

L2VPN KPI	Scale
Number of L2VPN Instances	8,000
Number of Interfaces	8,000

1.2 Metro L2VPN Topology

Test topology consists of three routers, PE1, P1 and PE2, where ACX7100-32C is the PE1 and ACX7100-48L is the PE2 and P1 is that of PTX10008. PEs are connected to CEs using 4 x 100G links. CE’s are simulated using a traffic generator. PEs are connected to the core using 8 x 100G links, which are bundled in 2 x LAGs each having four links.

The ACX7100-32C is the Device Under Test (DUT). The underlay transport used for testing is SR-MPLS with both OSPF and ISIS protocols individually. We also covered LDP as the underlay transport. The CE-bound interface configurations are of SP-Style (see glossary).



Test configures 8,000 L2VPN instances. Bidirectional traffic in iMIX mode at 99.9% offer-load flows for all the L2VPN services. A total of ~798 Gbps traffic flows through the DUT.

1.2.1 Base Configurations

```
regress@PE1> show configuration protocols |display inheritance no-comments
bgp {
    group IBGP {
        type internal;
        local-address 12.1.1.1;
        family inet {
            unicast;
        }
        family l2vpn {
            signaling;
        }
        neighbor 12.1.1.3 {
            export nhself;
        }
    }
}
mpls {
    interface ae0.0;
    interface ae1.0;
}
ospf {
    source-packet-routing {
        node-segment ipv4-index 101;
        srgb start-label 45000 index-range 2000;
    }
    area 0.0.0.0 {
        interface lo0.0 {
            passive;
        }
        interface ae0.0 {
            interface-type p2p;
        }
        interface ae1.0 {
            interface-type p2p;
        }
    }
}

regress@PE1>
```

Routing-Instance configuration required to bring up the L2VPN instances is as shown below.

```
interfaces {
    et-0/0/0 {
        flexible-vlan-tagging;
        encapsulation flexible-ethernet-services;
        unit 1 {
            encapsulation vlan-ccc;
            vlan-id 1;
```

```
        }
    }
}

routing-instances {
   METRO_L2VPN_VRF_1 {
      instance-type l2vpn;
      protocols {
         l2vpn {
            site s1 {
               interface et-0/0/0.1 {
                  remote-site-id 2001;
               }
               site-identifier 1001;
            }
            encapsulation-type ethernet-vlan;
            no-control-word;
         }
      }
      interface et-0/0/0.1;
      route-distinguisher 12.1.1.1:1;
      vrf-target target:65000:1;
   }

}
```

1.2.2 CE Bound Interface Schema

Each of the PE devices is connected to the traffic generator using 4 x 100G interfaces, each interface is logically split into 2000 VLANs to have 8000 interfaces, one interface bound to one L2VPN instance.

PE1	PE2	Port	Service Type	VRF	VLAN Association	Devices	Traffic Load
et-0/0/0	et-0/0/48:0	P1	L2VPN	1-2000	“1-2000”	4,000	99.9%
et-0/0/1	et-0/0/48:1	P2	L2VPN	2001-4000	“1-2000”	4,000	99.9%
et-0/0/2	et-0/0/48:2	P3	L2VPN	4001-6000	“1-2000”	4,000	99.9%
et-0/0/3	et-0/0/48:3	P4	L2VPN	6001-8000	“1-2000”	4,000	99.9%

1.2.3 L2VPN Verification

The first step to verify the L2VPN service is to make sure that the PE devices are having BGP established with l2vpn-signalling. As shown below from PE1, BGP is in Established state for Peer PE2(12.1.1.3).

```
regress@PE1> show bgp summary
Threading mode: BGP I/O
Default eBGP mode: advertise - accept, receive - accept
Groups: 1 Peers: 1 Down peers: 0
Table Tot Paths Act Paths Suppressed History Damp State Pending
inet.0
                 35 1 0 0 0 0
bgp.l2vpn.0
                8000 8000 0 0 0 0
Peer AS InPkt OutPkt OutQ Flaps Last Up/Dwn State|#Active/Received/Accepted/Damped...
12.1.1.3 65000 8024 8023 0 0 7:05 Establ
  inet.0: 1/35/35/0
  bgp.l2vpn.0: 8000/8000/8000/0
  METRO_L2VPN_VRF_1.l2vpn.0: 1/1/1/0
  METRO_L2VPN_VRF_2.l2vpn.0: 1/1/1/0
  METRO_L2VPN_VRF_3.l2vpn.0: 1/1/1/0
  ...
  METRO_L2VPN_VRF_7998.l2vpn.0: 1/1/1/0
  METRO_L2VPN_VRF_7999.l2vpn.0: 1/1/1/0
```

```
  METRO_L2VPN_VRF_8000.l2vpn.0: 1/1/1/0

regress@PE1>
```

The BGP l2vpn table of the METRO_L2VPN_VRF_1 instance is as follows:

```
regress@PE1> show route table METRO_L2VPN_VRF_1.l2vpn.0

METRO_L2VPN_VRF_1.l2vpn.0: 2 destinations, 2 routes (2 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

12.1.1.1:1:1001:2001/96
                *[L2VPN/170/-101] 00:09:40, metric2 1
                   Indirect
12.1.1.3:1:2001:1001/96
                *[BGP/170] 00:08:27, localpref 100, from 12.1.1.3
                  AS path: I, validation-state: unverified
                   to 15.1.1.2 via ae0.0, Push 45103
                 > to 15.1.2.2 via ae1.0, Push 45103

regress@PE1>
```

The L2VPN connections is UP towards the 12.1.1.3 Peer for the routing-instance METRO_L2VPN_VRF_1

```
regress@PE1> show l2vpn connections instance METRO_L2VPN_VRF_1
Layer-2 VPN connections:

Legend for connection status (St)
EI -- encapsulation invalid NC -- interface encapsulation not CCC/TCC/VPLS
EM -- encapsulation mismatch WE -- interface and instance encaps not same
VC-Dn -- Virtual circuit down NP -- interface hardware not present
CM -- control-word mismatch -> -- only outbound connection is up
CN -- circuit not provisioned <- -- only inbound connection is up
OR -- out of range Up -- operational
OL -- no outgoing label Dn -- down
LD -- local site signaled down CF -- call admission control failure
RD -- remote site signaled down SC -- local and remote site ID collision
LN -- local site not designated LM -- local site ID not minimum designated
RN -- remote site not designated RM -- remote site ID not minimum designated
XX -- unknown connection status IL -- no incoming label
MM -- MTU mismatch MI -- Mesh-Group ID not available
BK -- Backup connection ST -- Standby connection
PF -- Profile parse failure PB -- Profile busy
RS -- remote site standby SN -- Static Neighbor
LB -- Local site not best-site RB -- Remote site not best-site
VM -- VLAN ID mismatch HS -- Hot-standby Connection

Legend for interface status
Up -- operational
Dn -- down

Instance: METRO_L2VPN_VRF_1
Edge protection: Not-Primary
  Local site: s1 (1001)
    connection-site Type St Time last up # Up trans
    2001 rmt Up Oct 23 23:43:12 2022 1
      Remote PE: 12.1.1.3, Negotiated control-word: No
      Incoming label: 16, Outgoing label: 16
      Local interface: et-0/0/0.1, Status: Up, Encapsulation: VLAN
      Flow Label Transmit: No, Flow Label Receive: No

regress@PE1>
```

The following output captures the number of instances in the DUT

```
regress@PE1> show l2vpn connections |match "rmt Up" | count
Count: 8000 lines

regress@PE1> show route table bgp.l2vpn.0 |match BGP | count
Count: 8001 lines

regress@PE1> show route summary |match L2VPN: |count
Count: 16001 lines

regress@PE1> show route table mpls.0 protocol l2vpn |match L2VPN|count
Count: 16000 lines
```

1.2.4 Traffic Verification at Router at PE1

Traffic got load-balanced across the lag bundles ae0 and ae1 towards the core and across each of the member interfaces.

```
PE1 Seconds: 8 Time: 03:23:34

Interface Link Input packets (pps) Output packets (pps)

 ae0 Up 3125232690044 (31603996) 3092758811605 (31275639)
 ae1 Up 3060284770339 (30951593) 3092758802956 (31279892)
 esi Up 0 (0) 0 (0)
 et-0/0/0 Up 1546379263196 (15637224) 1546379158447 (15636310)
 et-0/0/1 Up 1546379214713 (15637071) 1546379163592 (15636440)
 et-0/0/2 Up 1546379149169 (15636320) 1546379164104 (15636866)
 et-0/0/3 Up 1546379157074 (15636913) 1546379157376 (15636834)
 et-0/0/4 Up 0 (0) 3568 (0)
 et-0/0/5 Down 0 (0) 0 (0)
 et-0/0/6 Up 796385380100 (8053366) 770096960274 (7787501)
 et-0/0/7 Up 777055626584 (7857916) 776282456360 (7850117)
 et-0/0/8 Up 770096919918 (7787568) 770096939235 (7787592)
 et-0/0/9 Up 781694763442 (7905146) 776282455736 (7850429)
 et-0/0/10 Up 770096934324 (7788487) 776282470328 (7850981)
 et-0/0/11 Up 763138220225 (7718340) 770096938714 (7788677)
 et-0/0/12 Up 761591833934 (7702754) 776282455310 (7851325)
 et-0/0/13 Up 765457781856 (7742012) 770096938604 (7788909)
```

1.3 Generating Scale Configuration

The Python Script is having three components, a Jinja File capturing the various configuration templates, a Params File capturing the various user variables and the script uses both these input files and generates the required configurations and commit that in the router.

Note: The configs are generated as per the CE-bound interface schema.

1.3.1 Jinja Configuration Template

The template contains configurations for both interfaces which are part of the routing instances and routing instance configurations.

```
# Routing Instance Configuration
# File Name: l2vpn_jinja.j2
# Version: 1.0
groups {{group_name}} {
   interfaces {
      {{ifd_name}} {
         flexible-vlan-tagging;
         encapsulation flexible-ethernet-services;
         {%- for unit_id, vlan_id in ifl_variables %}
         unit {{unit_id}} {
            vlan-id {{vlan_id}};
            encapsulation vlan-ccc;
         }
         {%- endfor %}
      }
   }
   routing-instances {
      {%- for vrf_id, ifl_name in l2vpn_variables %}
      METRO_L2VPN_VRF_{{vrf_id}} {
         instance-type l2vpn;
         interface {{ifl_name}};
         route-distinguisher {{router_id}}:{{vrf_id}};
         vrf-target target:{{local_as_no}}:{{vrf_id}};
         protocols {
            l2vpn {
               site s1 {
                  interface {{ifl_name}} {
```

```
                        remote-site-id {{remote_site_id}};
                    }
                    site-identifier {{local_site_id}};
                }
                encapsulation-type ethernet-vlan;
                no-control-word;
            }
        }
    }
    {%- endfor %}
  }
}
```

1.3.2 Configuration Parameters

‘‘‘



DC – AI – Routing – Switching – Security – Wireless – Automation

© Copyright 2026 Hewlett Packard Enterprise Development LP. The information contained herein is subject to change without notice. The only warranties for Hewlett Packard Enterprise products and services are set forth in the express warranty statements accompanying such products and services. Nothing herein should be construed as constituting an additional warranty.
Hewlett Packard Enterprise shall not be liable for technical or editorial errors or omissions contained herein.
Trademark acknowledgments, if needed. All third-party marks are the property of their respective owners.

HEWLETT PACKARD ENTERPRISE

HPE.com



